Stock Price Prediction with ARIMA, LSTM, and GRU

A Comparative Study on AAPL, MSFT, and NVDA

Foundation of Artificial Intelligence
Group Project Presentation

Dataset
FAANG stock prices with technical indicators

Tickers
AAPL · MSFT · NVDA

Team Members
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Sequence models versus a classical baseline — under the same forecasting setting.

Group Work Allocation

Member	Responsibilities
ZHOU Hang	Opening remarks, project framing, background and research questions
ZUO Jian	Dataset selection, data description and preprocessing pipeline
LIN Jiajun	Model introduction, methodology and experiment settings
QI Minghao	Results presentation, model comparison and performance analysis
HE Jingxi	Discussion, conclusion, future work

Background and Motivation

Why this project?

- Stock prediction is a typical **time-series** forecasting problem.
- It connects directly to course topics such as sequence modeling and deep learning.
- Public financial datasets make model comparison practical and reproducible.
- We focus on **forecasting performance**, not real trading profit.

Project framing

1. **Time-Series Data**
   Historical prices and technical indicators
2. **Model Comparison**
   ARIMA vs LSTM vs GRU
3. **Forecast Evaluation**
   RMSE, MAE, and MAPE

Key motivation

We wanted a project where the data pipeline, classical baseline, deep learning models, and evaluation metrics can all be shown clearly in one coherent workflow.

Research Questions

We focused on three questions:

Q1 Can ARIMA, LSTM, and GRU predict next-day closing prices effectively?

Q2 **Which model performs better on AAPL, MSFT, and NVDA under the same experimental setup?**

Q3 How does stock volatility affect **forecasting performance** and **model stability**?

*These questions structure the rest of our presentation: **data, methods, results, and discussion**.*

Dataset Description

FAANG stock prices with technical indicators

Tickers
AAPL MSFT
NVDA

Rows / ticker
2,494

Date range
2016-02-23
to
2026-01-22

Ticker	Rows	Date Range
AAPL	2494	2016-02-23 to 2026-01-22
MSFT	2494	2016-02-23 to 2026-01-22
NVDA	2494	2016-02-23 to 2026-01-22

Why these three stocks?
They share the same data structure and date range, but differ in price dynamics and volatility. That makes them suitable for fair comparison and discussion.

Feature groups

- **Price & volume** — Open, High, Low, Close, Volume
- **Moving averages** — SMA, EMA
- **Momentum signals** — RSI, MACD, MACD Signal
- **Bands & risk** — Bollinger Bands, Daily Return, Volatility

EDA and Data Preprocessing

DATA

Preprocessing pipeline

1 Raw CSV
Multi-ticker stock data

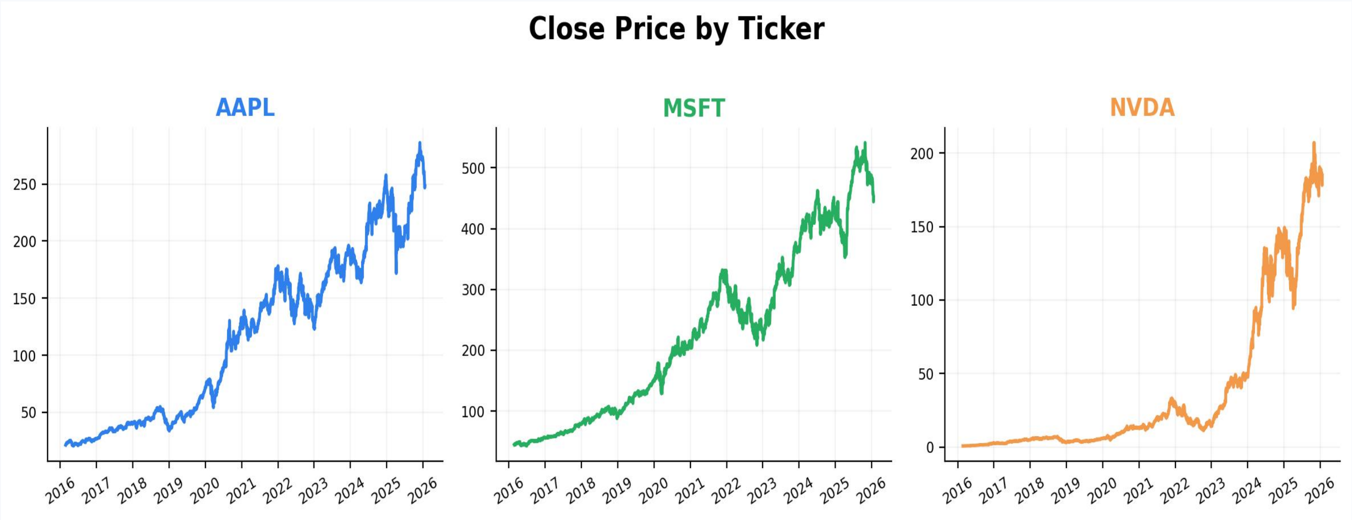
2 Filter by ticker
AAPL / MSFT / NVDA

3 Clean data
Sort dates + fill missing values

4 30-day sequences
Build sliding-window samples

5 Chronological split
Train / Val / Test

EDA snapshot



Key preparation decisions

- Window size = 30 trading days
- Chronological split: 70% / 15% / 15%
- Sequences per ticker = 2,465
- Excluded leakage-prone future-price columns
- Kept the same split across all models

Methods

Model comparison under the same forecasting task

ARIMA
Classical statistical baseline

• Designed for time-series forecasting
• Good at short-term dependency modeling
• Used here as a strong, interpretable benchmark

LSTM
Deep learning sequence model

• Uses memory cells for sequential patterns
• Can model complex temporal signals
• Often benefits from larger data and tuning

GRU
Simpler recurrent neural network

• Similar goal to LSTM with fewer gates
• Often easier to train
• Used here as a lighter deep learning alternative

Task Next-day closing price prediction

Metrics RMSE · MAE · MAPE

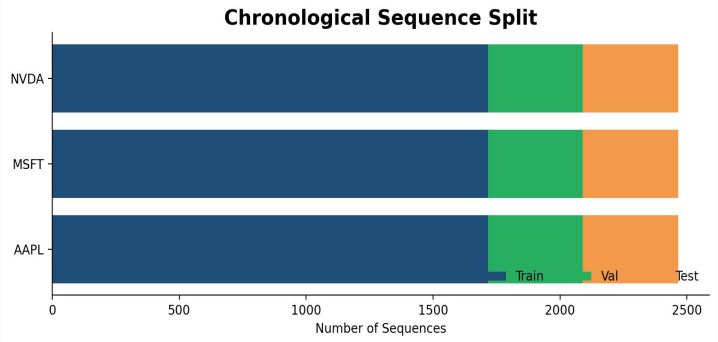
Goal Fairly compare a strong baseline against two sequence models

Experiment Settings

To keep the comparison fair, we fixed the same data setup across models.

Setting	Value
Tickers	AAPL, MSFT, NVDA
Window size	30 days
Target	Next-day closing price
Split	Chronological 70 / 15 / 15
Feature count	16 input features
Test samples / ticker	375

LSTM / GRU hyperparameter	Value
Hidden size	64
Number of layers	2
Dropout	0.2
Learning rate	0.001
Batch size	32
Patience	8



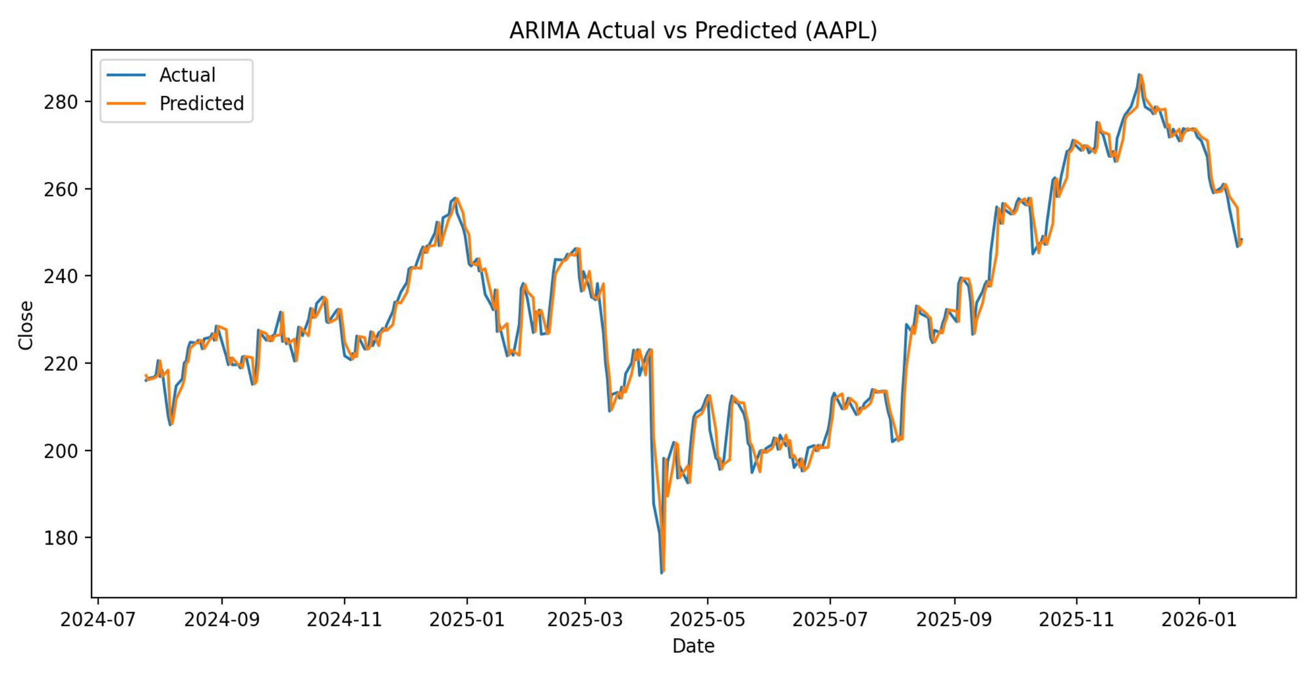
Fair comparison principle

All models used the same tickers, the same 30-day window, the same chronological split, and the same test sets. This keeps the comparison interpretable and fair.

ARIMA Results

ARIMA delivered the strongest and most stable baseline across all three stocks.

Ticker	Best order	RMSE	MAE	MAPE
AAPL	(3,1,0)	3.89	2.60	**1.16%**
MSFT	(5,1,0)	6.26	4.43	**1.00%**
NVDA	(3,1,0)	4.15	3.10	**2.24%**



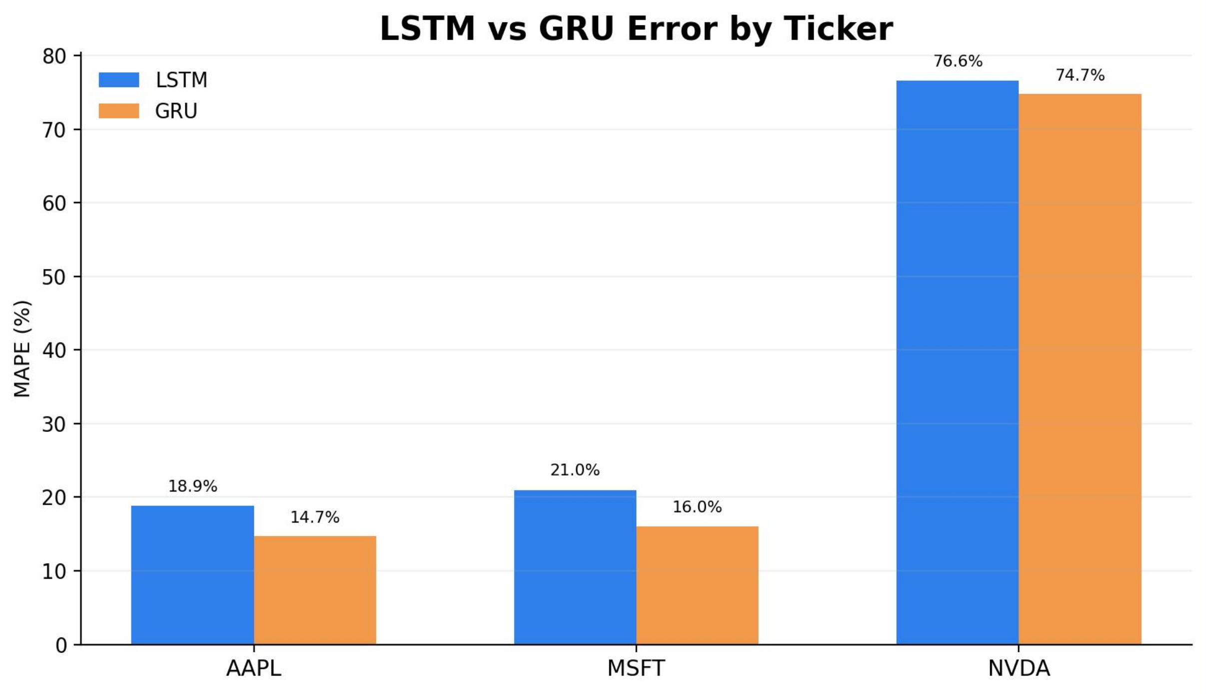
Key takeaway

- MAPE stayed around 1%–2% on all three stocks.
- Best orders were simple: (3,1,0) or (5,1,0).
- This makes ARIMA a very strong baseline for short-term forecasting.

LSTM and GRU Results

Deep learning models did not outperform ARIMA in our current setup.

Ticker	Model	RMSE	MAE	MAPE
AAPL	LSTM	49.66	45.28	18.86%
AAPL	GRU	39.57	35.40	**14.68%**
MSFT	LSTM	105.57	97.56	20.98%
MSFT	GRU	82.82	74.84	**15.99%**
NVDA	LSTM	118.04	114.48	76.62%
NVDA	GRU	115.48	111.82	**74.74%**



Reading the result

GRU was slightly better than LSTM on AAPL and MSFT, but both models struggled on NVDA, where the error became very large.

Discussion

Findings

- ARIMA was stronger in this one-step forecasting task.
- Deep learning did not beat the classical baseline here.
- NVDA was much harder to predict than AAPL and MSFT.
- Lower loss did not automatically mean better test performance.

Possible reasons

- One-step stock forecasting may already favor a strong short-term statistical model.
- Each ticker had only about 1,716 training sequences, which limits deep learning capacity.
- Stock prices are noisy and non-stationary, and higher volatility likely made NVDA harder to model accurately.
- Better training fit did not necessarily lead to better generalization on unseen test data.

Deep learning is not always better under every task setting.

Conclusion and Future Work

Main takeaways

1 Complete pipeline

We built a full pipeline from preprocessing to model comparison.

2 Strong baseline

ARIMA gave the best and most stable forecasting performance.

3 Clear insight

LSTM and GRU did not outperform ARIMA in our current setup.

4 Future directions

Use more data, predict returns, and add external signals.

Future work

- use more data more effectively
- predict returns instead of raw prices
- add external signals such as news sentiment
- tune hyperparameters

Thank you!